MIL-T-11838D
13 September 1972
SUPERSEDING
MIL-T-11838C
20 September 1963

MILITARY SPECIFICATION

TREADWAY, SUPERSTRUCTURE: BRIDGE,

FLOATING; FOOT, ALUMINUM

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the treadway of the floating aluminum footbridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-E-527	- Enamel, Alkyd, Lusterless.

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

FSC 5420

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Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

ME

TA13211E8560 - Treadway, Superstructure: Bridge, Floating; Foot, Aluminum.
D13207E9958 - Treadway, Superstructure: Bridge, Floating, Foot, Aluminum, Packing of.

(Copies of specification, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The treadway shall be as shown on Top Assembly TA13211E8560 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one treadway for examination within the time frame specified (see 6.2), to prove prior to starting production, that his production methods will produce treadways that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Identification marking. The treadway shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.5 Treatment and painting. All parts of the treadway shall be cleaned, treated, and painted as specified on the drawings except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly. The raised portions of the entire top of the corrugated deck shall be surfaced, not less than 72 hours after painting, with one coat of nonslip material conforming to MIL-W-5044, type II, color number 34087. Following painting and surfacing, white paint conforming to TT-E-527, composition G, shall be applied as follows:

(a) A stripe running the length of the corrugated deck on the centerline, 3 inches wide, which may be applied to the raised portions only.
(b) The exposed upper surface of the upper handrail brackets.
(c) The machined surfaces in the throat of female connector plates from the end to a point 2 inches from the end.
(d) The inner exposed web surfaces of the girders (find numbers 1 and 2) outlined by the end of the girder and the vertical centerline of the retainer lug.

3.6 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each treadway. The letters "NET WT" shall precede the

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numerals in the weight and the letters "CU" shall precede the cubic displacement. The markings shall be Gothic-type capitals and Arabic numerals, 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.7 Workmanship.

3.7.1 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid excessive scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches, or winds. Corners shall be square and true. Aluminum 3/8 inch thick and over shall be cut by routing or sawing. Aluminum under 3/8 inch thick may be cut by shearing. Aluminum shall not be cut by methods using heat.

3.7.2 Riveted connections. A jig or frame shall be employed in making riveted connections to insure alinement and maintain required tolerances. Assemblies bolted prior to riveting shall show no undue separation. Rivets shall be driven cold with pressure tools, where practicable, or with a pneumatic hammer. The driven head, except as otherwise specified on the drawings, shall be button or cone point when formed with a squeeze riveter or pneumatic hammer or squash or flathead when formed by a bucking bar or dolly. The countersunk heads for locator plates may be driven or manufactured. Rivet heads shall be full, concentric with the rivet shank axis, and shall have full bearing on the surface of the member.

3.7.3 Welding. The surfaces of parts to be welded shall be free from scale, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loadings. Welds shall be free from such defects as cracks, flux inclusions, spatter, undercutting, porosity, lack of fusion with parent metal, or incorrect size.

3.7.4 Interchangeability. Treadways shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Acceptability criteria. Treadways which conform to all requirements in sections 3 and 5 of this specification and pass all examinations in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction treadway shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each treadway shall be inspected for conformance to the dimensions covered by the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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4.5 Inspection procedure.

4.5.1 Examination. The treadway shall be examined as specified herein for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. Riveting not as specified.
107. Nonslip covering not applied to corrugated panels as specified.

Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking incorrect, missing or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.
205. Bolts not peened or staked as specified on drawings.

4.6 Inspection of preparation for delivery. The packing and marking shall be examined to determine compliance with section 5 of this specification. Any packing that is not in accordance with the drawing or any marking that is missing, illegible, incorrect, or incomplete shall constitute failure and shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. The treadways shall be packed in accordance with Drawing D1320TE9958.

5.1.2 Level C. The treadways shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.2 Marking. Bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The treadway is intended for use as the deck section of the floating aluminum footbridge.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production treadways from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish treadways conforming to this specification.

6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 International standardization. Certain provisions of this specification are the subject of International Standardization Agreement ABC, Standard 88. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:
Army - ME

Preparing activity:
Army - ME

User activity:
Navy - MC

Project No. 5420-0093

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-11838D, Treadway, Superstructure; Bridge, Floating; Foot, Aluminum

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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